Academic Stress and Suicide Risk Among South Korean Adolescents

By

PAKEEZA ABID

THESIS

Submitted to

KDI School of Public Policy and Management

In Partial Fulfillment of the Requirement

For the Degree of

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Committee in charge:

Professor David Sungho Park, Supervisor ____________________

Professor Yoon Chungeun ____________________

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Abstract

This study examines the impact of the start of the academic year, March 2nd on suicide deaths in Korean adolescents, aged 11 to 19. I exploit difference-in-difference method using data from Korea Microdata Integrated Service (MDIS) and Korean Statistical Information System (KOSIS). The treatment group consists of the individuals aged 11 to 19 ”students”, and the control group includes all other individuals in the dataset. I find a significant increase in suicide deaths among Korean students, aged 11 to 19 following March 2nd and the increase is not significant in population other than students. The result suggests that the start of the academic year in Korea on March 2nd every year triggers the increase in suicide rates among students.

Keywords: Suicide, Adolescents, South Korea, Mental Health, Academic Pressure

Contents

1 Introduction

1.1 Overview of Suicide Trends in South Korea

Suicide has been South Korea's most critical public health issue for more than two decades. Although suicide is recognized as a major global issue, South Korea's suicide rate has consistently remained higher than most of the developed countries. The number of suicides per every 100,000 South Koreans, went up to 29.1 in 2025, the highest level since 2011. This level remains substantially above the OECD average, 2.4 times the OECD average of 10.8, and it shows that the suicide rate in South Korea is due to deep rooted social and societal factors rather than just short-term fluctuations.

Although suicide affects individuals of all ages, but its impact is particularly alarming among adolescents and young adults, with new government data showing that intentional self-harm has overtaken cancer and has become the top cause of death among people between their 10s-40s for the first time since 1983.

Adolescence in Korea is characterized by intense academic pressure, highly competitive society, and strong expectations encompassing educational attainments and future success. These factors may increase psychological distress during a life stage already marked by major emotional and cognitive changes.

1.2 Academic Stress and Adolescent Vulnerability

The South Korean educational system is widely known for its rigorous academic requirements and intense competition. According to the Organization for Economic Cooperation and Development (OECD), South Korean students study an average of 49.43 hours per week, which is 15 hours longer than the global average. High stake examinations like CSAT represents the intensity of the system and these examinations can serve as primary contributor to academic stress in South Korean adolescents. These conditions may contribute to elevated levels of stress among adolescents, increasing the risk of mental health challenges such as depression, anxiety, and suicidal ideation. Previous studies have shown how academic pressure plays a significant part in shaping Korean students' mental health. Therefore, it is crucial to explore how different aspects of academic system interact with the suicide risk.

1.3 Institutional Context: South Korean Academic Calendar and Structural Pressures

One unique aspect of the South Korean education system is the nationally coordinated academic calendar. Every year, on March 2 (the starting date of the academic term), primary, middle and high school students across the country go back to school for a new academic year.

Beginning the academic year is a huge institutional shift for students. Students step into new classrooms, meet new teachers and take on updated academic expectations during this period. For many students, the transition comes with shifts in peer groups and changes in school atmospheres.

On top of these social adjustments, the start of the school year brings with it a new cycle of academic evaluation and performance pressure. Anxiety overload when it comes to academic expectations and performance outcomes in the future for students. These pressures can be even more pronounced at the beginning of the school year due to the huge societal pressure placed on educational success.

The start of the academic year happens on a set nationally coordinated date that helps create an institutional barrier that will likely affect student behaviour and psychological strain levels. This institutional setting offers an opportunity to assess whether suicide outcomes are systematically patterned around the codetermined academic year.

1.4 Problem Statement and Policy Context

Suicide remains a serious problem in South Korea, and the government has tried to address it through various policies over the years. The government launched and implemented the first 5-year National Strategy for Suicide Prevention, led by the Ministry of Health and Welfare in 2004. It was the first major push to work together at the national level to raise public awareness, develop prevention programs, and make mental health services more accessible to all. In 2009, the second 5-year National Strategy for Suicide Prevention was re-established and in 2012, the Korea Suicide Prevention Center was established, this center took on the task of uniting national efforts and coming up with proven ways to reach people before it was too late. In 2013, they promoted “Suicide CARE”, a Korean standardized suicide prevention program for gatekeeper intervention. Despite these extensive policy efforts and substantial public investment in suicide prevention programs, the suicide rate in South Korea remains high. This shows that while good things are being done, current methods may be missing some important parts. There may be issues related to the structure of society, such as pressures at school or work, family expectations or attitudes to mental health, which are not fully addressed by existing policies. It is crucial to understand the ongoing factors, such as the school system and work pressures, related to suicide risk. By understanding how specific settings affect people, the government can design effective policiesand can provide timely support and reach those who need it most.

1.5 Research Gap and Motivation

Many researchers have studied suicide rate of South Korea, but much of the existing literature focuses on general socioeconomic and psychological determinants like depression, unemployment, social isolation and income inequality. Research examining suicide risk in youth frequently highlight the role of mental health challenges, academic pressure and other societal defects. However, most of this literature relies primarily on survey-based evidence or cross-sectional analysis that perceived stress or psychological disorders.

Relatively few of the researchers have examined the temporal distribution of suicide in relation to the complex features of academic system, such as the structural constraints and entrenched pressures within academic environments. While existing literature links academic pressure and social expectations to adolescent mental health, studies largely focus on internal psychological factors. There is a need to shift focus toward longitudinal trends of suicide incidence within educational settings, moving beyond static psychological determinants to analyze temporal changes in risk.

Understanding whether suicide risk increases during specific institutional transitions is important for both academic research and public policy. This study addresses this research gap by analyzing whether suicide rate among the students exhibits a systematic spike around a specific time of the year. By combining administrative mortality data with the institutional timing of the school calendar, the analysis provides new evidence on how educational transitions may influence suicide risk among adolescents.

1.6 Research Objectives

This study aims to answer one primary research question, whether there is a correlation between the beginning of the school year in South Korea and changes in the rates of suicide incidents for students. In other words, the research aims to determine whether there is an increase in the rates of suicide incidents for students compared to non-students during the beginning of the school year.

The research aims to accomplish three objectives. The first objective of this research is to identify and explore the temporal patterns of suicide by analyzing the data to determine whether there is specific spike in suicide rates during certain periods of the year. The second objective of this research is to assess whether there are variations in the temporal patterns of suicide incidence between different population groups, particularly between students and non-students.

And the third and the main objective of this research is to explore whether there is a corelation between the start of the academic year and suicide rate in students.

By addressing these objectives, the study aims to provide new empirical insights into how institutional features of the education system may influence suicide risk among adolescents.

In particular, the study aims to examine if the beginning of the school year generates academic pressures and anxiety that may lead to the incidence of suicide among students. The basic research question that this study intends to examine is: Does the opening of the school year heighten the risks of suicide among Korean students? In particular, the study aims to examine the following: Does the beginning of the academic year heighten anxiety and academic pressures that may lead to the incidence of suicide? Descriptive study results indicate that the suicide rate peaks on March 2, which is the beginning of the academic year in Korea, but only among students.

1.7 Contribution of the Study

This study contributes to the literature on suicide risk and adolescent mental health in several important ways. First, it provides new empirical evidence on the temporal concentration of suicide risk in relation to institutional events within the education system. While previous studies have examined the relationship between academic stress and mental health outcomes, relatively little research has analyzed how suicide incidence varies around specific points in the academic calendar.

Second, the study utilizes large-scale administrative mortality data, allowing for the analysis of suicide patterns at a highly disaggregated temporal level. This approach helps to identify short-term spikes in suicide incidence that may not be visible in annual or aggregated statistics.

Third, by employing a Difference-in-Differences empirical framework, the study isolates changes in suicide incidence among students relative to non-students around the start of the academic year. This design allows the analysis to exploit institutional timing as a source of variation in suicide risk.

Lastly, the findings are applicable to policy-making in that if the risk of suicide among students is higher at certain periods, such as at the beginning of the school term, suicide prevention policies can be formulated to target this period, and this could make the current policies more effective in suicide prevention in the country.

2 Literature Review

2.1 Academic Stress and Adolescent Mental Health

Adolescence has been identified as a critical developmental stage characterized by rapid emotional and social changes during which academic pressure has been recognized to have a significant effect on adolescents' mental health status. Academic pressure has been identified globally as one of the key factors that have been linked to increased levels of mental health disorders in adolescents, including heightened levels of anxiety and depression. Empirical research has supported academic pressure as one of the key factors that have been linked to suicidal behaviors in adolescents. In the study Depression and Suicidal Ideation as a Consequence of Academic Stress among Adolescent Students (Nandagaon & Raddi, 2020) found that academic stress is linked to depression among adolescents. Standardized psychological evaluations were carried out on 1,204 adolescents in India to determine whether academic pressure was linked to suicidal ideation in adolescents. The study found that academic pressure was significantly and positively correlated with depression and suicidal ideation in adolescents in India. Similarly, in the study Academic Stress and Suicidal Ideation (Okechukwu et al., 2022) carried out in southern Nigeria with 505 university students, academic pressure was found to be closely linked to suicidal ideation in adolescents. Similar trends have been noted in South Korea through various research done in the region, where academic and social pressure were found as significant factors contributing to the youth suicide rate. In research on Adolescent Suicide in South Korea (Kwak & Ickovics, 2019), This study highlights that academic stress, depression, and social stigma drive high adolescent suicide rates in South Korea, recommending multi-dimensional interventions involving schools, peers, and parents to promote resilience and mental health. Country must develop and implement interventions to improve mental health and reduce suicide, especially among vulnerable children and adolescents. However, a different perspective was noted in Suicide Trends Among South Korean Young Adults, where a study of death statistics (H. Choi et al., 2025) between 2013 and 2020 noted that for young adults in South Korea between 20 and 39 years of age, economic factors affected male suicides, while female suicides in the region were affected more by relationship factors. Collectively, these studies underscore the significant impact of academic stress on the mental health and suicidal tendencies of adolescents and young adults, thereby underscoring the need for an urgent intervention at the policy level.

2.2 Academic Stress and Suicidal Ideation: International Evidence

However, recent international research has sought to identify the different processes through which academic stress can influence suicidal thoughts or behaviors. The focus of these studies is on finding out the correlation between academic stress, societal expectations, and mental health.

A study by Jarvis et al. (2020) focuses on the relationship that exists between academic stress and social capital in the South Korean education system. The findings of this research reveal that the academic stress that exists in the South Korean education system can result from the fact that the South Korean education system is very competitive. Other studies also investigate the role of school and family environments on adolescent suicide risk. J. Park et al. (2025) conducted a study on how school and parental environments affect suicidality among South Korean youth. The study concluded that academic pressures, along with parental and school-related pressures, both directly and indirectly contribute to suicidality among school-going youth. This study also emphasizes that academic pressures are part of a social context.

Though this literature offers valuable information on the mental processes that connect academic pressures and suicidality, it is observed that most of these studies are based on survey data and concentrate on individual mental processes instead of population-level suicide mortality rates.

2.3 Suicide Risk Among Adolescents in South Korea

Suicide has now become one of the most prominent causes of death for teenagers in South Korea. Various researchers have studied the demographic and social factors that influence the risk of suicide.

In the study by S.-U. Lee et al. (2018), the authors analyze the long-term trends in the rates of suicide in South Korea from the year 1993 to 2016. The authors note that suicide rates have increased over the years. The study suggests that socioeconomic factors, demographic factors, and social pressures have all contributed to the increase in the rates of suicide for different demographic groups.

Other researchers have also studied the influence of environmental factors on the risk of suicide. In the recent study by Jang et al. (2022), the authors analyze the relationship between the rates of suicide and the characteristics of the social environment in different regions of South Korea. Various factors that influence the risk of suicide have been identified. These include socioeconomic factors, demographic factors, urbanization, and health behaviors.

Studies on adolescents emphasize the need to address academic and social pressure. According to Kwak and Ickovics (2019), academic stress, mental health stigma, and lack of access to mental health services have been identified as factors that contribute to the high rate of suicidal behavior in Korean adolescents. The authors stress the need for multi-dimensional approaches to prevent suicidal behavior in this population.

In another study, S. Shin and Kim (2023) employ machine learning to predict suicidal ideation in Korean children and adolescents. Their study validated psychological distress, depression and academic stress as strongest predictors of suicidal ideation (AUC > 0.85).

2.4 Structural and Institutional Risk Factors for Suicide

Apart from the individual-level psychological factors, recent studies have also emphasized the importance of the structural/institutional environment. These include the influence of social expectations, labor market pressures, and institutional systems on mental health. Y. G. Kim et al. (2016) carried out a comparative study on the differences in the patterns of suicidal behavior in South Korea and other OECD countries. The findings of the study indicated that the rate of suicidal behavior in South Korea was higher than the OECD average. The authors of the study argued that the structural factors play an important role in the differences in the suicidal rates. The institutional environment, such as the education system, may also have an influence on suicidal behavior. The education system of South Korea is considered to be extremely competitive. This competitive education system has often been considered one of the factors that may have an influence on the stress level of teenagers. Academic performance is considered an important determinant of socioeconomic status. Therefore, students have to perform well in academics. The pressures may become more prominent at the beginning of the school year or during the time of examinations.

2.5 Temporal Patterns in Suicide Risk: Existing Evidence

While there is a plethora of research on the long-term patterns and socio-economic factors that contribute to suicide, there is a relative scarcity in research on the temporal patterns of suicide incidence in a given calendar year. Understanding the timing of when there is a higher incidence of suicide can provide valuable insight into the potential factors that might contribute to a suicide event in a given institution or society. Previous research on the temporal patterns of suicide has provided evidence that there are seasonal patterns in the incidence of suicide in a number of countries. However, there is a relative scarcity in research that examines the potential relationship between suicide and specific institutional factors, such as the transition between school levels. While there is a plethora of research on academic stress, there is a relative scarcity in research that examines the potential patterns in suicide mortality that might coincide with the start of the academic calendar year.

2.6 Limitations of Existing Literature

Despite substantial advances in the understanding of the factors associated with suicide risk, there remain some gaps within the literature.

First, much of the literature on academic stress makes use of data obtained from surveys, which often utilize perceived stress or psychological symptoms as measures of stress. Although such literature offers valuable insights into the experiences of stress among individuals, it may not be representative of the experiences of the larger population.

Second, much of the literature on suicide outcomes measures such events on a yearly basis, which may not be representative of short-term changes within suicide incidence rates.

Third, there is limited literature on the analysis of suicide risk, which makes use of high-frequency data on mortality rates, which can be used for detailed analysis of temporal changes. The relationship between institutional transitions, such as the beginning of the academic year, and suicide incidence rates has not been explored extensively.

2.7 Conceptual Framework

This study contributes to existing literature on the relationship between academic stress and adolescent mental health and suicide risk. The theoretical background for the analysis is based on a conceptual model that suggests that changes in the educational system could be a stress event that could affect suicide risk. In particular, the beginning of a new academic year could create a stressful situation for students in terms of new expectations in terms of their academic performance and their social life. Such a stressful situation could increase anxiety levels in students, which in turn could increase the risk of suicide during a specific period. Therefore, this research paper examines whether there is an increase in the incidence of suicide in students during the beginning of a new academic year compared to non-students.

3 Data and Methodology

3.1 Data Sources and Description

National Suicide Mortality Data

This study utilizes mortality data obtained from South Korea and focuses on exploring the patterns of suicide deaths across demographic and temporal dimensions. The data comprises 3.5 million records of deaths, each providing information on demographic and contextual characteristics of the deceased individuals. The data contains variables such as age, gender, education level, occupation, region, place of death, and time of death.

The data type is pooled cross-sectional data, and it from year 1997 to year 2023. This type of data can be used for the analysis of demographic data and time variation in the data on the mortality of individuals who committed suicide.

The data type is characterized by the availability of data on the time of death, which can be used for the analysis of the trends of suicide at the daily level. This can help identify any short-term increases in the incidence of suicide that may have occurred on account of institutional or social events.

The data is mainly obtained from two official statistical platforms maintained directly by Government of South Korea: The Microdata Integrated Service (MDIS) and Korean Statistical Information Service (KOSIS), operated by Ministry of Data and Statistics.

The Microdata Integrated Service provides researchers with access to detailed micro-level datasets derived from official national statistics, including mortality records compiled by Statistics Korea. These datasets contain anonymized individual-level information on deaths recorded across the country.

Occupation Classification and Occupation Code 13 (Students and Unemployed)

The occupation status is available in the mortality data set in standardized categories. For this study, occupation code 13 is of particular importance, which comprises of students and the unemployed only.

The analysis of the differences in the suicide outcomes between the occupation code 13 and other occupation codes allows for the identification of whether there are any variations in relation to students and non-students at the beginning of the school year.

Age Classification and Adolescent Population Definition (11–19 Years)

The age at the time of death is noted in years and is used to determine patterns in death by suicide that are specific to certain ages. For this study, specific attention has been given to persons aged 11 to 19 years, which constitutes the adolescent group, people in elementary, middle and high School, which are most likely to be affected by academic-related stressors. By focusing specifically on this group, it becomes possible to determine whether patterns in death by suicide among adolescents coincide with the onset of the academic year.

3.2 Descriptive Statistics

These statistics offer insights into how the death rate due to suicides varies with age groups, gender, and occupation categories. This analysis can be used to determine what groups in society have the highest number of death rates due to suicides.

Apart from statistics, exploratory data analysis methods are used to analyze trends in death rates due to suicides over time. These methods involve trend analysis and visualization techniques like drawing trend lines and time-series plots to determine if there are any trends or spikes in death rates due to suicides during any time of the year.

Table 1 presents the descriptive statistics for the key variables included in the dataset.

Table 1: Summary Statistics

Variable	N	Mean	SD
Age	323869	51.308	20.920
Female	323869	0.312	0.463
Married	323869	0.284	0.451
Student and Unemployed	323869	0.576	0.494
Year	323869	2011.352	7.230

3.3 Variable Construction

Mortality data includes a variable for the time of death, which allows for the aggregation of suicides on a daily basis, allowing for the analysis of short-term temporal patterns, including the possibility of peaks in suicide incidence, such as those associated with institutional events such as the beginning of the academic calendar. To analyze differences between various groups of the population, there is aggregation of suicides by occupation, particularly comparing those who fall into occupation code 13, which includes students and the unemployed, against other occupation codes. Furthermore, there is age-based aggregation of suicides, particularly for adolescents between the ages of 11 and 19, which allows for analysis of whether there are peaks in suicide incidence among school-aged adolescents, as well as whether the peaks are associated with student populations.

3.4 Empirical Strategy

Difference-in-Differences Framework

This study examines whether the start of the academic year is directly associated with changes in suicide rate among students in South Korea. The empirical strategy exploits the institutional timing of the South Korean academic calendar, where the school year begins nationally on March 2. This date represents a sharp transition for students and provides a natural point around which suicide outcomes can be compared.

The analysis compares suicide rate between students, which are classified by occupation code 13 and ages between 11 to 19 (Treatment Group) and the rest of the population, which serve as the control group.

The duration of the treatment was determined by the start of the academic year, which occurs on March 2 in South Korea. To account for the short-term effects of the transition into the new academic term, I set the window of observation to 20 days, consisting of 10 days before and 10 days after March 2. In the context of the model, the time after March 2 represents the treatment time, whereas the 10 days before March 2 representthe pre-treatment time. The basic idea behind the empirical strategy is to analyze the change in the number of suicides among students around the onset of the school year in relation to the change in the number of suicides among non-students in the same time period.

Empirical Stratergy

To estimate the effect of the beginning of the academic year on suicide incidence, the following Difference-in-Differences specification is estimated:

$$Y_{it} = \alpha + \beta(Treat_i \times Post_t) + \gamma_r + \theta_o + \lambda_y + \mu_m + \epsilon_{it} \tag{1}$$

In the empirical specification, Y_{it} represents the suicide outcome for group i at time t. In the empirical analysis, the outcome variable corresponds to suicide deaths aggregated at the monthly level.

$Treat_i$ is an indicator variable equal to one if the observation belongs to the student group (occupation code 13), and zero otherwise. $Post_t$ is a time indicator equal to one for observations occurring on or after March 2, which marks the beginning of the academic year, and zero for observations prior to this date. The interaction term $Treat_i \times Post_t$ captures the differential change in suicide outcomes among students following the start of the academic year.

The empirical models progressively incorporate additional fixed effects to control for potential sources of unobserved heterogeneity. Specifically, the regressions include region fixed effects (γ_r), occupation fixed effects (θ_o), year fixed effects (λ_y), and month fixed effects (μ_m). These fixed effects account for time-invariant differences across regions and occupations as well as common temporal shocks and seasonal patterns affecting suicide rates.

Standard errors are clustered at the regional level to account for potential correlation in suicide outcomes within regions over time. The coefficient of interest is β, which measures the change in suicide outcomes among students after the beginning of the academic year relative to non-students.

Parallel Trends

Identification requires that suicide outcomes for students and non-students would have evolved similarly in the absence of the academic-year transition. This assumption is assessed by examining suicide trends for both groups during the pre-treatment period prior to March 2.

Visual inspection of pre-treatment trends indicates that suicide outcomes among students and non-students follow similar patterns before the beginning of the academic year. This similarity in trends provides support for the parallel trends assumption underlying the empirical strategy.

Following the start of the academic year, however, suicide outcomes among students diverge from those of the comparison group, suggesting that the institutional transition associated with school reopening may be linked to increased suicide risk among students.

4 Empirical Results

4.1 Descriptive Trends in Suicide Rates in South Korea

In the following section, descriptive evidence on the patterns of suicides in South Korea will be provided before the actual causal analysis. Investigating the long-term trends in the incidence of suicides is essential to understand the temporal patterns of the phenomenon in the country. First, the evolution of the number of suicides in the country from 1997 to 2023 will be examined, followed by the investigation of the patterns in the number of suicides on a monthly basis.

Annual Suicide Trends (1997–2023)

Figure 1 illustrates the annual trend of suicide deaths in South Korea, covering the period from 1997 to 2023.

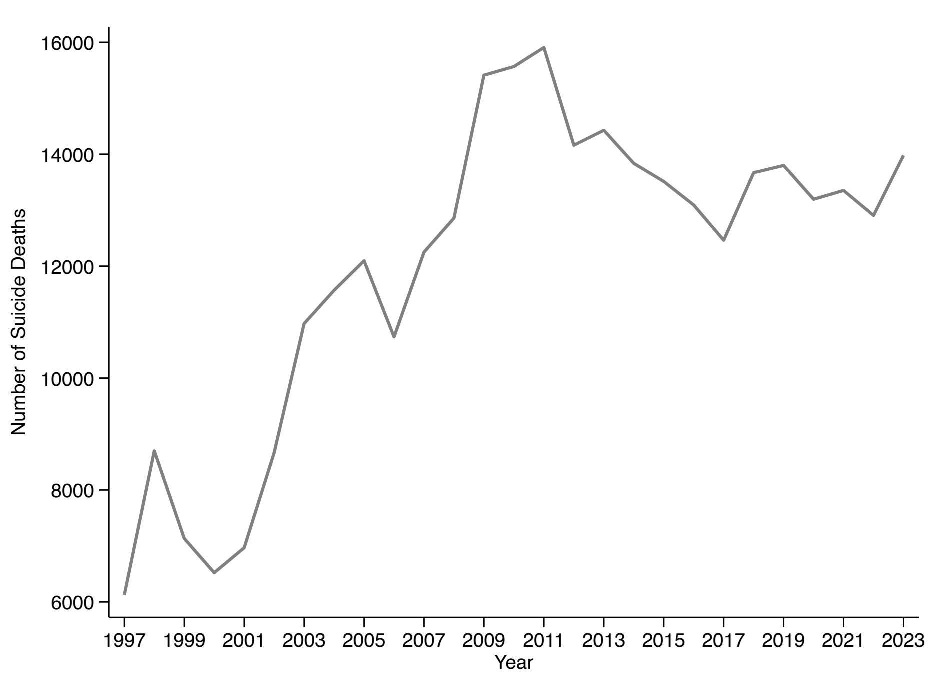

Figure 1: Annual Suicide Trends in South Korea (1997–2023)

From Figure 1, it is evident that there is a significant fluctuation in the data from 1997 to 2023. There is a notable increase in suicide deaths during the early 2000s and a remarkable spike during 2008 and 2009, a period when the global financial crisis occurred. After this spike, deaths resulting from suicide peaked during the early 2010s before gradually declining over the years. There are notable decreases during 2012 and 2017 before it again start to fluctuate and remain high.

However, despite some reductions during certain years, the overall trend reveals that the incidence of suicide has been a constant public health issue in South Korea over the past two decades. Though the annual data shows the overall pattern of suicide rate in South Korea, it does not reveal the pattern in the incidence of suicides throughout the calendar year. To analyze the pattern on micro level, the analysis now turns to the monthly pattern of suicides.

Monthly Distribution of Suicide Deaths

To examine the pattern of seasonal variations in suicide deaths, the number of suicide deaths is aggregated on a monthly basis for all the years in the data set. This pattern of average suicide incidence in the months of the year is shown in Figure 2.

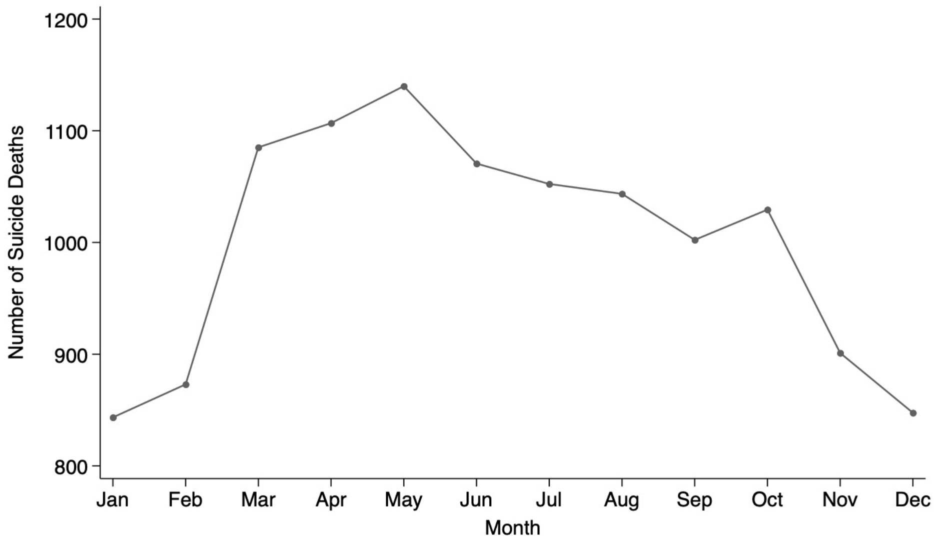

Figure 2: Monthly Distribution of Suicide Deaths in South Korea

Note: The figure shows the average number of suicide deaths for each calendar month in South Korea. Monthly values are calculated by first aggregating suicide deaths within each year–month and then averaging these monthly totals across all years in the sample period.

As shown in the figure 2, the distribution of suicide deaths on a monthly basis reveals that it does not remain uniform throughout the year. The number of suicide deaths steadily increases from the beginning of the year and peaks during the months of spring, especially in March. After this month, the incidence of suicide deaths steadily decreases during the remaining months of the year. This pattern during the months of the year reveals that the number of suicides could be influenced by certain factors, especially those that change during the months of the year. However, to determine the day-to-day variations in suicide incidence, the following sections will examine the suicide deaths on a day-to-day basis.

4.2 Occupation Distribution of Suicide Deaths

By understanding the distribution of suicides in terms of occupation categories, it is possible to gain a deeper understanding of the population groups to which the highest incidence of suicide deaths can be attributed. Occupation categories reflect various important aspects of the socioeconomic status and life stages of individuals and, therefore, can highlight variations in suicide deaths in relation to various groups of the population.

Table 2 highlights the distribution of suicide deaths in relation to various occupation categories.

Table 2: Tabulation of Occupation Code

Occupation Code	Occupation Name	Freq.	Percent	Cum.
1	Parliamentary officers, senior officers and managers	4315	1.36	1.36
2	Professionals	11234	3.54	4.90
3	Technicians and associate professionals	17835	5.62	10.51
4	Office workers	4848	1.53	12.04
5	Service and sales workers	27129	8.54	20.58
6	Skilled agricultural, forestry and fisheries workers	23668	7.45	28.03
7	Artisans and related skilled workers	8038	2.53	30.56
8	Machine operators and assemblers	6310	1.99	32.55
9	Simple labor workers	17836	5.62	38.16
13	Students and unemployed	186614	58.75	96.92
99	Unknown, military personnel (excluding enlisted personnel)	9796	3.08	100.00
Total		317623	100.00	

From the distribution, it is evident that there is a wide range of variation in the incidence of suicide among different occupation groups. Although it is evident that there are cases of suicide deaths across a wide range of occupation groups, it is also evident that the largest proportion of deaths is accounted for by individuals whose occupation is categorized as 13, which includes students and the unemployed. This category accounts for 57.6 percent of the total deaths by suicide.

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4.3 Age Distribution of Suicide Deaths

The suicide rates also differ between age groups. Hence, it is crucial to understand which groups contribute most to suicide deaths. Using this data, the suicides are grouped based on age and analyzed to understand how suicide risk is distributed across an individual's lifecycle. Although suicides happen in all age groups, in this particular study, emphasis is placed on adolescents between the ages of 11 and 19 years because they are the ones most exposed to their environment in an institution of learning. This is because this age group in South Korea is mostly in middle and high school. They are exposed to various pressures in their lives that may affect their mental health in one way or another. Being in an institution of learning means that their suicides can be affected by events in their environment, such as when school starts on March 2. The following section examines temporal patterns in suicide incidence, with particular attention to daily fluctuations in suicide deaths across the calendar year.

4.4 Student vs. Non-Student Suicide Trends

As the rate of suicide deaths can change over time, it is essential to determine if there is a change in the rate of suicide over particular periods of the year. By using the data on the time of death contained in the national mortality data set, the rates of suicide incidents were aggregated on a daily basis to identify high-frequency patterns. This data has been aggregated over all years contained in the data set, 1997 to 2023, to identifyrecurring patterns and reduce yearly fluctuations.

Figure 3 shows monthly trends of suicide rates for the comparative population group. This figure offers an overview of how rates of suicide change over the course of a calendar year for a general population group, during which monthly trends tend to be constant with moderate fluctuations.

Figure 4 shows monthly trends of suicide rates for individuals classified under occupation type 13, only between age 11 to 19, identified as students. By comparing these two figures, it is possible to determine if there is a change in the rate of suicide incidents between students and a general population group, particularly during early March when the academic year begins in South Korea.

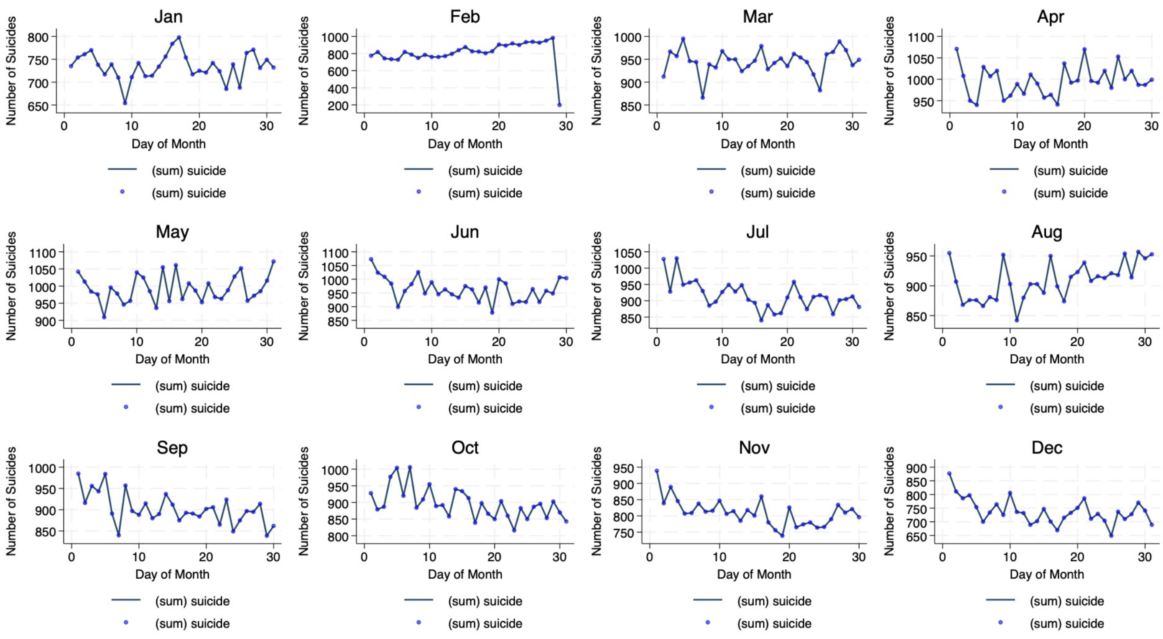

Figure 3: Monthly Distribution of Suicide Deaths in South Korea

Note: This figure presents the monthly distribution of suicide deaths among all the individuals (Control Group) except the treatment group, individuals aged 11 to 19 ”students” in South Korea.

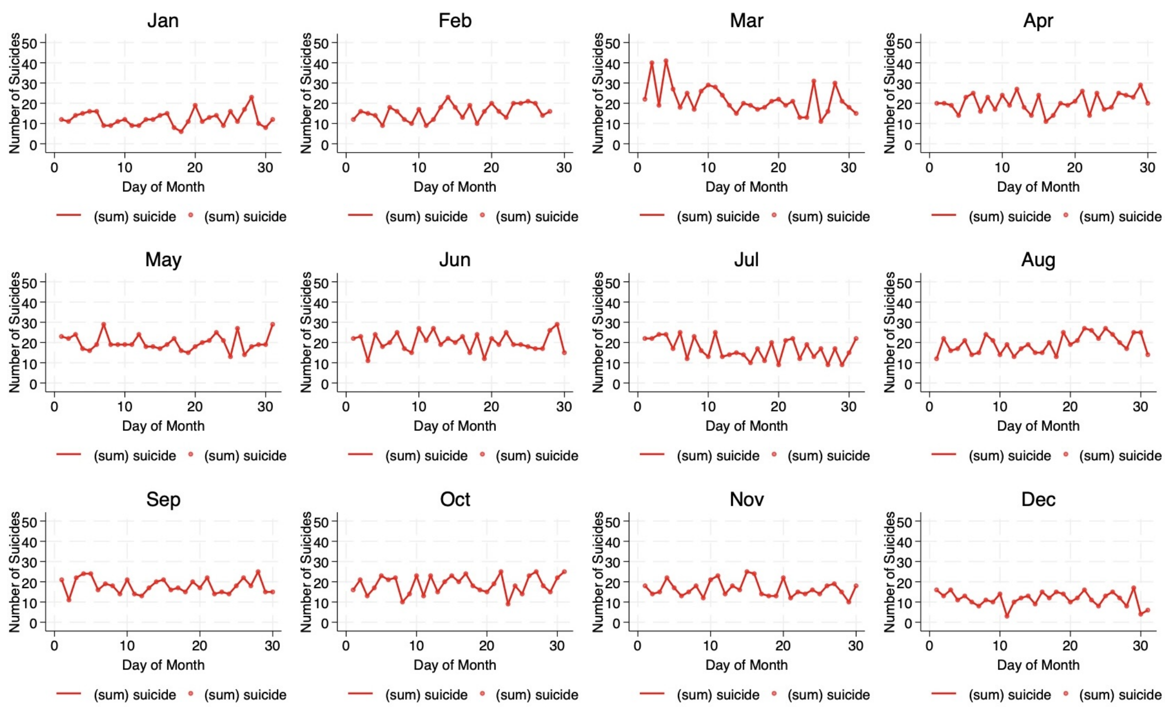

Figure 4: Monthly Distribution of Suicide Deaths in South Korea

Note: This figure presents the monthly distribution of suicide deaths among individuals aged 11 to 19 ”students” (Treatment Group) in South Korea.

This pattern may indicate that the transition associated with the beginning of the school year may be related to an increase in suicide risk for students. The reason for this is that the school calendar is synchronized across the nation, making this date a universal transition for students across the nation.

4.5 Comparative Graphical Analysis

Figure 5 indicates the daily trends of suicides across the calendar year for the whole population on a micro-level. There is a certain degree of seasonality in the data, and it can be observed that while there are some fluctuations in the rates of suicides on a daily basis.

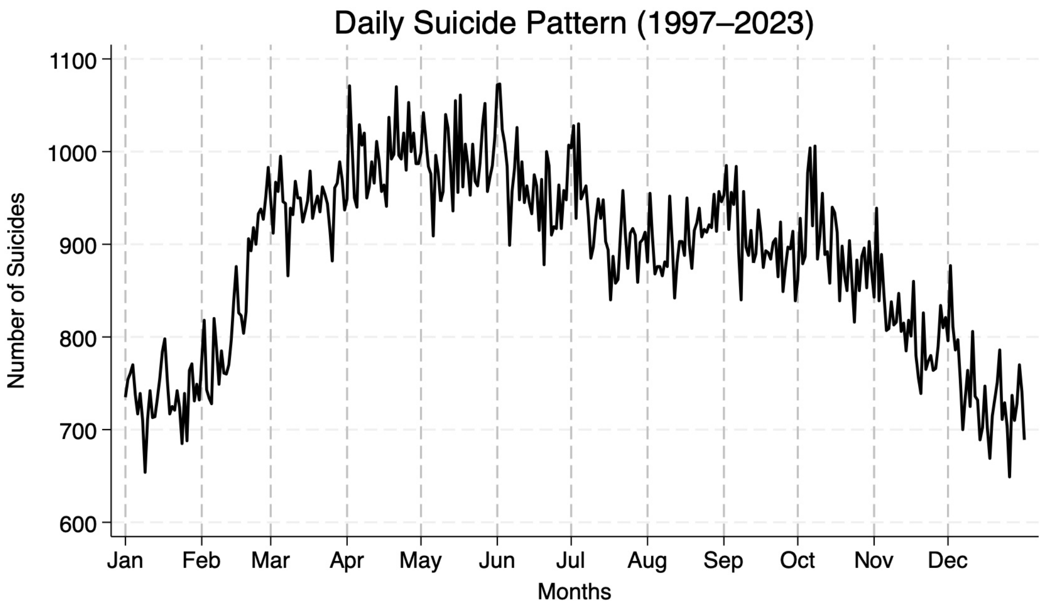

Figure 5: Monthly Distribution of Suicide Deaths in South Korea

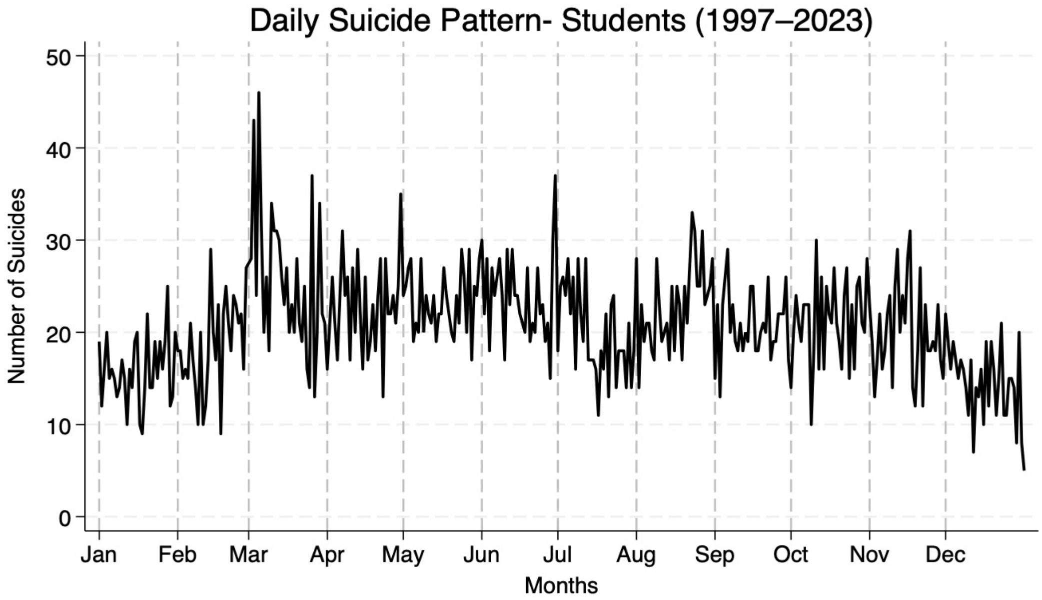

Figure 6: Monthly Distribution of Suicide Deaths in South Korea

Figure 6 indicates the daily trends of suicides for students aged 11 to 19 years. By comparing this figure with Figure 5, it can be observed that while there are higher rates of fluctuation on a daily basis, certain periods of the year experience higher rates of fluctuation than others. Most prominently, there is an observable increase in rates of suicides during early March, which coincides with the start of a new academic term in South Korea.

Evidence of Concentrated Risk Among Students

The comparison between the two groups offers evidence that the temporal spike in early March is likely concentrated among students. While the overall population experiences relatively stable patterns of daily variation, the student population experiences an increase in the incidence of suicide at the beginning of the academic year. The academic calendar is synchronized nationally, meaning that the start of the school year is an event that occurs simultaneously for students across the country. Therefore, the start of the academic year represents a transition that students collectively experience. The increase in the incidence of suicide at this time suggests that transitions related to the academic calendar may coincide with periods of high psychological stress for students. The graphical evidence above does not prove causality but offers an important form of support for the hypothesis that the incidence of suicide increases for students at the time of the year when the academic calendar begins. These findings inform the empirical approach used in this study, which tests whether the start of the academic calendar represents a change in the incidence of suicide for students relative to non-students.

4.6 Regression Results

To formally evaluate whether the beginning of the academic year is associated with changes in suicide incidence among students, a Difference-in-Differences regression model is used. The treatment group consists of students aged 11–19 classified under occupation code 13, while the control group includes non-students. The post-treatment period is defined as dates occurring on or after March 2, the official start of the academic year in South Korea.

Table 3: Difference-in-Differences Regression Results

	(1) Model 1	(2) Model 2	(3) Model 3	(4) Model 4
Treat X Post	0.285^{***}	0.294^{***}	0.294^{***}	0.294^{***}
	(0.0366)	(0.0363)	(0.0365)	(0.0365)
Region FE	Yes	Yes	Yes	Yes
Occupation FE	No	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes
Month FE	No	No	Yes	Yes
Observations	447156	435690	435690	435690

Standard errors in parentheses

Standard errors are clustered at the regional and age levels.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

An event study analysis is conducted to explore the timing and dynamics of the treatment effect further. The event study specification estimates the effect of start of academic year on suicide incidence for each day relative to March 2, allowing a visual check for pre-treatment trends and evolution in treatment effect after beginning of academic year.

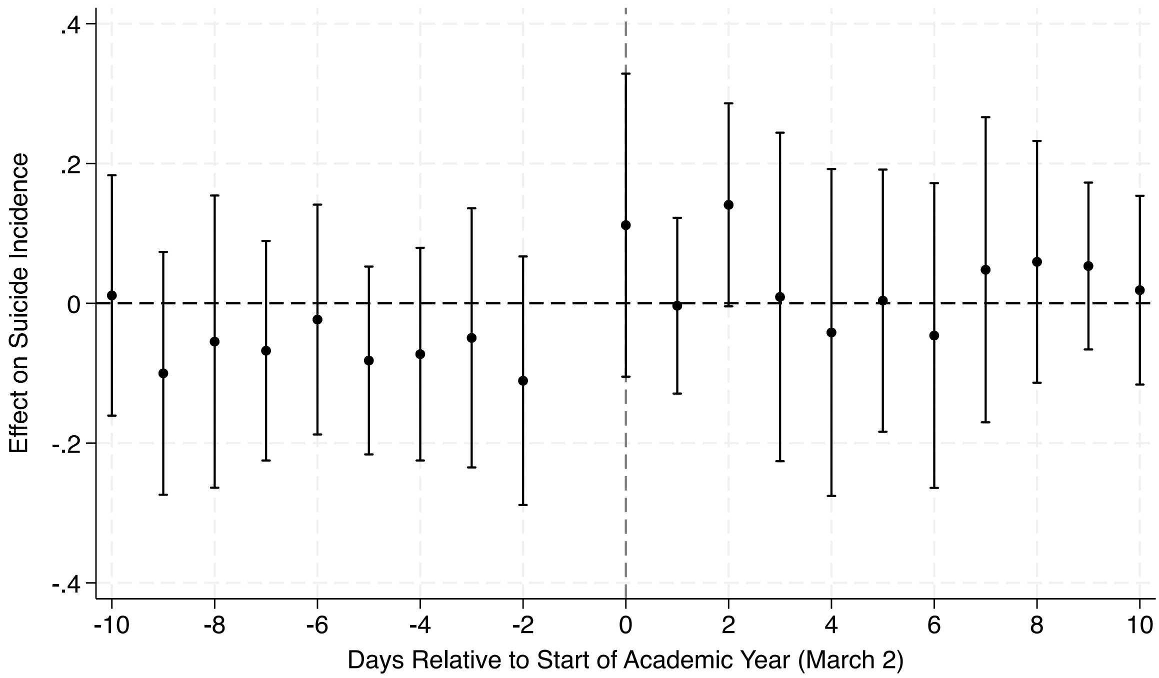

Figure 7: Event Study of Suicide Incidence Around the Start of the Academic Year

Note: The figure plots event-study estimates of the impact of the start of the academic year (March 2) on suicide incidence among students relative to non-students. Coefficients represent differences relative to the day preceding the start of the academic year. Vertical bars indicate 95% confidence intervals. Standard errors are clustered at the regional and age levels.

The estimated coefficient on the days just prior to the start of an academic year are near zero and statistically indistinguishable from zero (see figure 7), supporting evidence consistent with the parallel trends assumption. After March 2 the coefficients are positive indicating a higher incidence of suicides among students compared to non-students after the academic year began. This pattern is consistent with the difference-in-differences estimates shown in Table 3.

4.7 Weekly Suicide Patterns Among Students

Aside from the monthly and daily variations, the incidence of suicide deaths of students also varies from day to day of the week. Figure 7 illustrates the distribution of suicide deaths for individuals aged 11-19 and occupation code 13 for the seven days of the week. The pattern reveals that the incidence of suicide deaths peaks on Mondays and gradually decreases towards the end of the week, reaching its lowest point on Saturdays before increasing again on Sundays. This could be explained by the psychological pressure of the beginning of the school week, where students have to go back to their academic routines and responsibilities after the weekend. While this pattern of suicide deaths on a weekly basis is just illustrative, it again points to how the academic week could affect the incidence of suicide deaths of students.

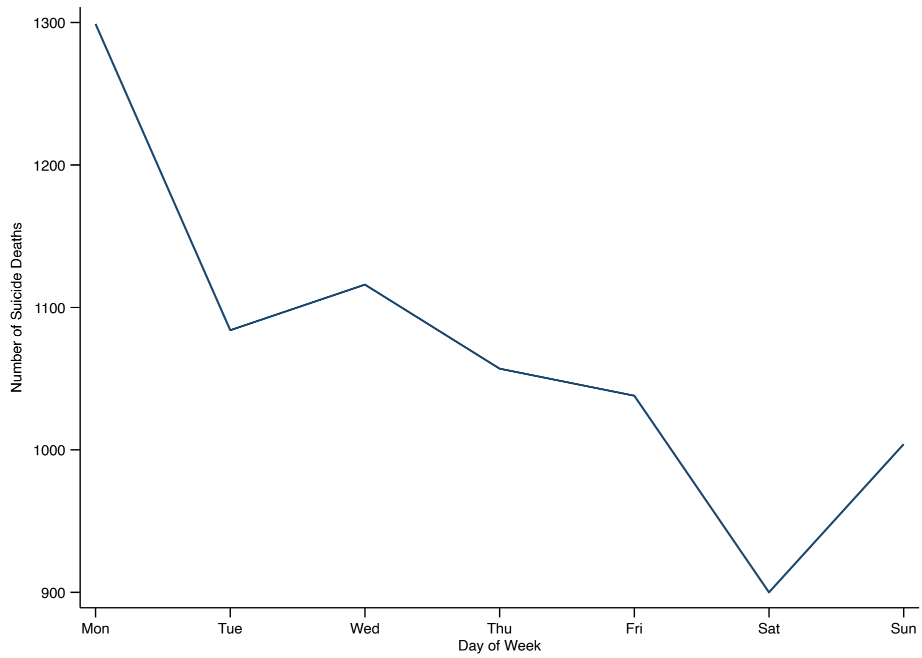

Figure 8: Weekly Suicide Patterns Among Students

5 Discussion

5.1 Interpretation of Findings

From the empirical analysis, there are a number of important patterns that are observed regarding the suicide rate in South Korea. First, a huge concentration of suicides is coming from the group classified under occupation type 13, which includes students and the unemployed. Secondly, the range of ages from 11 to 19 is a significant population when analyzing the results relating to the education system and the incidence of suicides. Thirdly, there is a significant rise in the incidence of suicides in early March, marking the beginning of the academic season. The most important observation is that this rise is more pronounced among students than in the general population. The results indicate that there is room for improvement in suicide prevention strategies, which can be achieved by taking into account specific periods of increased risk, rather than implementing general prevention strategies. The fact that the start of the school year is a predictable, nationally synchronized event makes it a potential opportunity for prevention strategies. Strategies can be implemented by the school system and education authorities during this period, such as early mental health screening, increased access to counseling, and strategies for adapting to school transitions, which may reduce psychological stress among vulnerable youth.

5.2 Limitations

This study has a few limitations. One such limitation is that occupation code 13 comprises both students and unemployed individuals. This limits the study's ability to perfectly identify students using occupation data alone. While it has been ensured that only individuals between 11 and 19 years of age are considered, a certain level of error can still exist. Moreover, the scope of the study is limited to establish a causal relationship between school reopening and suicide rates. While it has been observed that a certain pattern exists, it is possible that other factors could also play a part.. Lastly, it has been observed that the data does not include information on individuals' mental states or school environments, which could also be a reason for suicide.

6 Conclusion

This study analyzed the correlation between academic institutional transitions and suicide risk among young people in South Korea by analyzing national mortality data between 1997 and 2023. The results of the study indicate that there is a high concentration of deaths among individuals categorized under occupation code 13, which comprises students and the unemployed. Additionally, it indicates that young people between 11 and 19 years of age form a critical demographic category. The results of the temporal study on daily suicide rates indicate a significant spike on March 2, marking the start of the academic calendar in South Korea. The results indicate that this spike is more pronounced among students than other individuals. This is supported by the results of the Difference-in-Differences study, which indicates that there is a relative increase in suicide risk among students during the transition to a new academic year compared to other individuals.

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